

IT342 Phase 3: Web Development Completion Report

Project Name: BidWarsOnline

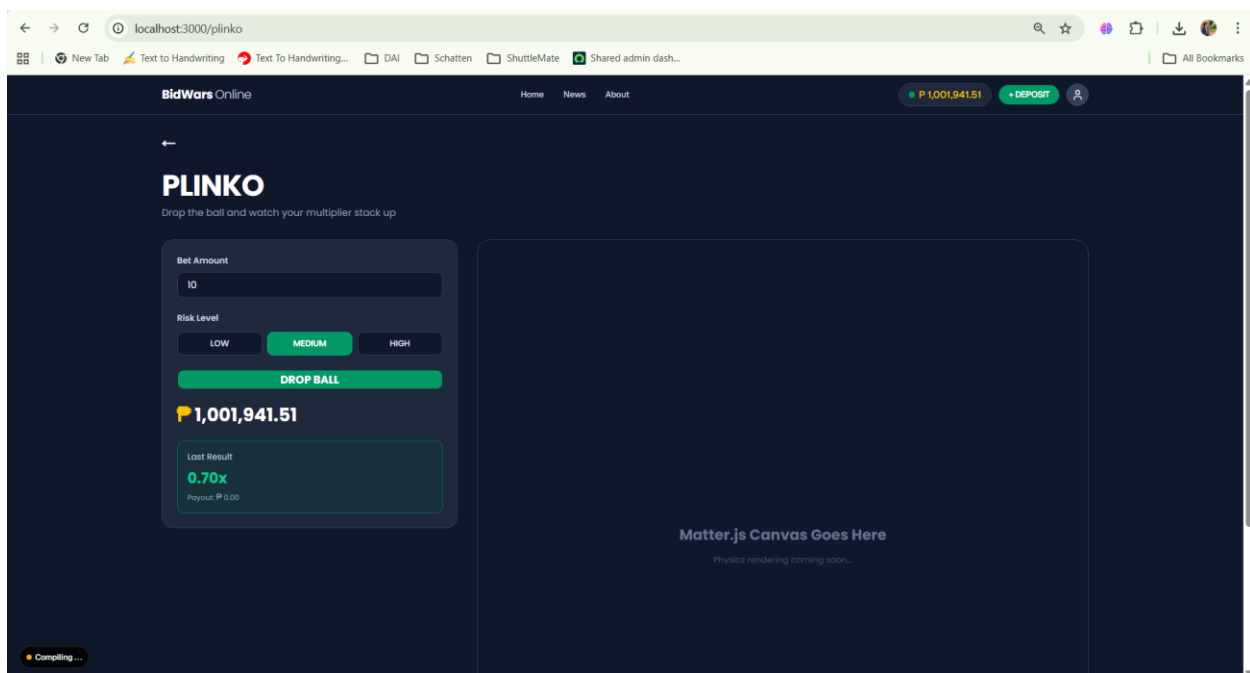
Student Name: Sanchez

1. Submission Requirements

- **GitHub Repository:** <https://github.com/Franz-Raven/IT342-Sanchez-BidWarsOnline>
 - **Final Commit Message:** IT342 Phase 3 – Web Development Completed
 - **Commit Hash:** [61a45e22e0194061a1b71969e22d95a1b936cbb2](#)
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2. Application Screenshots

A. Plinko Game



The user interface for playing the plinko game, matter.js UI onhold due to complexity.

id	bet_amount	created_at	game_data	game_type	multiplier	payout
106	10.0000	2026-04-12 10:25:53.290774	{"gameType":"PLINKO","risk":"MEDIUM",	PLINKO	0.7000	0.0000
105	10.0000	2026-04-12 10:12:19.785516	{"gameType":"MINES","minesCount":3,"g	MINES	1.9570	19.5700
104	10.0000	2026-04-12 10:08:00.30404	{"gameType":"HILO","result":"BUST","tim	HILO	0.0000	0.0000
103	10.0000	2026-04-12 09:41:52.50179	{"gameType":"HILO","streak":2,"timestam	HILO	4.3730	43.7300
102	10.0000	2026-04-12 09:36:01.957201	{"gameType":"HILO","result":"BUST","tim	HILO	0.0000	0.0000
101	10.0000	2026-04-12 09:34:58.98820	{"gameType":"HILO","result":"BUST","tim	HILO	0.0000	0.0000
100	10.0000	2026-04-12 09:34:23.31677	{"gameType":"HILO","streak":1,"timestam	HILO	1.2610	12.6100
99	10.0000	2026-04-12 09:34:04.72450	{"gameType":"HILO","result":"BUST","tim	HILO	0.0000	0.0000
98	10.0000	2026-04-12 09:26:29.72942	{"gameType":"HILO","result":"BUST","tim	HILO	0.0000	0.0000
97	10.0000	2026-04-12 09:19:20.245715	{"gameType":"HILO","result":"BUST","tim	HILO	0.0000	0.0000
96	10.0000	2026-04-12 09:18:43.66643	{"gameType":"HILO","result":"BUST","tim	HILO	0.0000	0.0000
95	10.0000	2026-04-12 08:59:19.489519	{"gameType":"HILO","timestamp":"2026-0	HILO	0.0000	0.0000
94	10.0000	2026-04-12 08:59:02.010182	{"gameType":"HILO","timestamp":"2026-0	HILO	0.0000	0.0000

Database record for the plinko game

3. Project Summary

How the Mines Game Works

The Mines feature is a grid-based risk game where users navigate a 5x5 board to uncover gems while avoiding hidden explosives. The process begins when the user sets their bet amount and selects the number of active mines via the Next.js/Kotlin interface. Upon starting the game, the Spring Boot backend securely generates a randomized grid, intentionally withholding the mine locations from the front-end to ensure provable fairness and prevent cheating. As the user clicks individual tiles, the system checks the hidden grid state. Finding a gem dynamically increases the potential payout multiplier based on combinatorial probability. Hitting a mine instantly busts the session, resulting in a loss. Users can manually trigger a "Cash Out" action after any successful click to instantly transfer their accumulated winnings to their digital wallet.

Inputs and Validations Used

The system captures the user's betAmount, the minesCount, and the specific tileIndex (0-24) during gameplay. The backend rigorously validates these inputs by ensuring the user has a sufficient wallet balance before deducting the bet , restricting the minesCount strictly between 1 and 24, and verifying that a submitted tileIndex has not already been clicked in the active session.